

Student ID:

1. You are given an unsorted array (a) containing both positive and negative integers. Write a Java program to determine the smallest missing positive integer in the array.

Sample Input	Sample Output
2, -3, 4, 1, 1, 7	3
5, 3, 2, 5, 1	4
-8, 0, -1, -4, -3	1

```

public class SetAB {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("How many numbers: ");
        int n = sc.nextInt();
        int[] a = new int[n];
        System.out.print("Enter the numbers: ");
        for (int i = 0; i < n; i++) {
            a[i] = sc.nextInt();
        }
        // sort the array first
        for (int i = 0; i < n - 1; i++) {
            int pos = i;
            for (int j = i + 1; j < n; j++) {
                if (a[j] < a[pos]) {
                    pos = j;
                }
            }
            if (i != pos) {
                int c = a[i];
                a[i] = a[pos];
                a[pos] = c;
            }
        }
        // the array is sorted now
        int missing = 1;
        for (int i = 0; i < a.length; i++) {
            if (a[i] == missing) {
                missing++;
            } else if (a[i] > missing) {
                break;
            }
        }
        System.out.println(missing);
        sc.close();
    }
}

```